

POWER SHELL PRACTICAL LAB MANUAL

Basic Commands, System Administration and Azure/Active Directory

PowerShell Basic Commands

Status: Open

This practical manual introduces Windows PowerShell and provides hands-on commands for basic PowerShell usage, objects and program flow, files and pipeline parameter binding, PowerShell providers and drives, WMI/CIM, networking and security monitoring, the PowerShell console and scripting environment, Azure administration, pipeline-specialized commands, and Active Directory/Microsoft Entra ID.

General Requirements

- Windows 10/11 or Windows Server with PowerShell available.
- Administrator privileges may be required for some system administration commands.
- For Azure exercises, an Azure account/subscription and appropriate permissions are required.
- For Active Directory exercises, a domain environment and RSAT/ActiveDirectory module may be required.
- For Microsoft Entra ID exercises, Microsoft Graph PowerShell and appropriate permissions are required.

01. Configuring Windows PowerShell and Working with Basic Commands

Aim

To configure the PowerShell environment and learn commonly used PowerShell commands for system administration.

1. Check PowerShell Version

```
$PSVersionTable
```

This displays the installed PowerShell version and environment information.

2. Display the Current User

```
$env:USERNAME
```

```
whoami
```

3. Display Computer Name

```
$env:COMPUTERNAME
```

4. Get Help

```
Get-Help Get-Process
```

```
Get-Help Get-Process -Examples
```

5. Find Commands

```
Get-Command
```

```
Get-Command *Process*
```

```
Get-Command *Service*
```

6. Get System Information

```
Get-ComputerInfo
Get-CimInstance Win32_OperatingSystem | Select-Object Caption, Version, OSArchitecture
```

7. Work with Files and Folders

```
Get-ChildItem
Get-ChildItem C:\
New-Item -ItemType Directory -Path .\PowerShellLab
Set-Location .\PowerShellLab
New-Item -ItemType File -Name test.txt
Get-Item test.txt
Remove-Item test.txt
```

8. Display Text

```
Write-Host "Welcome to PowerShell"
$name = Read-Host "Enter your name"
Write-Host "Hello $name"
```

9. Variables

```
$a = 10
$b = 20
$sum = $a + $b
$sum
```

10. Get Running Processes

```
Get-Process
Get-Process | Select-Object Name, Id, CPU
```

11. Get Windows Services

```
Get-Service
Get-Service | Where-Object {$_.Status -eq 'Running'}
```

12. Basic Network Commands

```
Get-NetIPConfiguration
Get-NetIPAddress
Get-NetAdapter
Test-Connection google.com -Count 4
```

13. Configure Execution Policy

```
Get-ExecutionPolicy
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Use execution-policy changes carefully. The CurrentUser scope avoids changing the policy for every user on the computer.

14. Create and Run a PowerShell Script

Create a file named hello.ps1 with the following contents:

```
$name = Read-Host "Enter your name"
Write-Host "Hello $name!"
```

Run the script from its folder:

```
.\hello.ps1
```

Exercise

- Display your PowerShell version.
- Display your computer name and current username.

- Create a folder named PowerShellLab and create a text file inside it.
- Display the list of running services.
- Find all commands containing the word Process.
- Test connectivity to a website using Test-Connection.

Viva Questions

- What is PowerShell?
- How is PowerShell different from Command Prompt?
- What is a cmdlet?
- What does Get-Help do?
- What is the purpose of Get-Command?
- What is an execution policy?
- What is the difference between a variable and an environment variable?